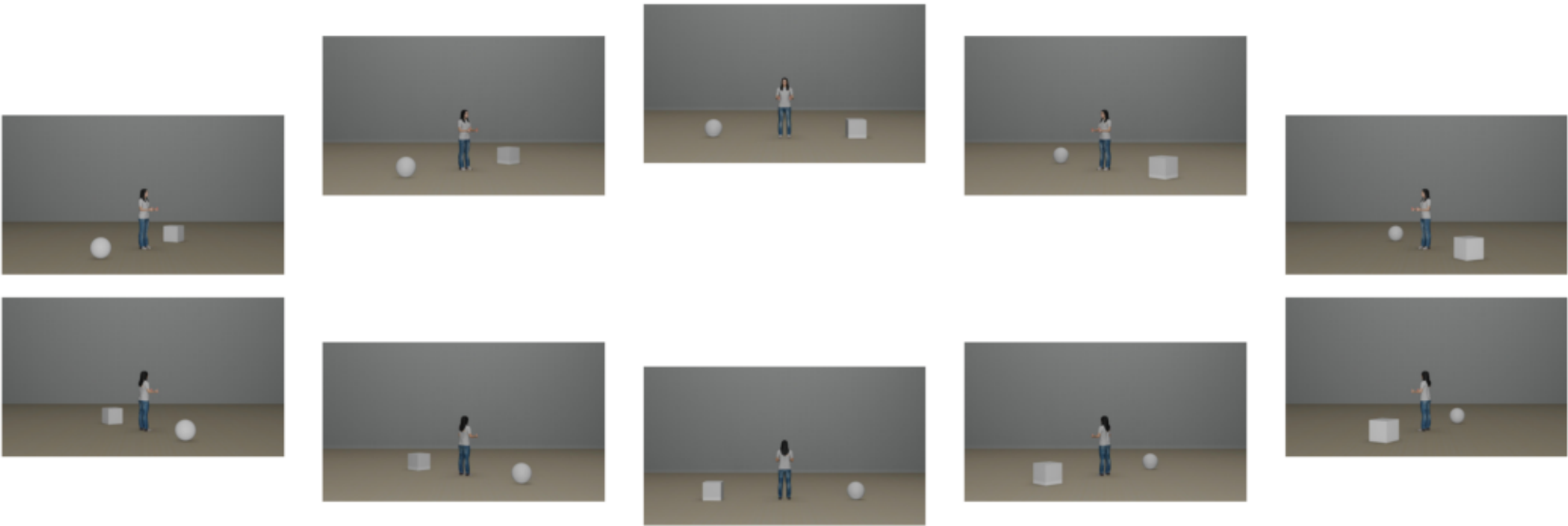


(a)



(b)

Embodiment Tokens



1. Extract keypoints

Diagram showing a person's head (LS) and shoulder (RS) keypoints. The distance between them is $\Delta x = 192 - 172 = 20$ and $\Delta y = 81 - 53 = 28$. The orientation is calculated as $\theta = \arctan2(-(28), 20) \times \frac{180}{\pi} + 360 = 305.55^\circ$.

2. Compute orientation

Token sequence:
<POSE_START> <SHOULDER_L>
<X_172> <Y_53> <SHOULDER_R> ...
<ORIENT_START> <YAW_6> <TORSO_W_2>
<ORIENT_END> <POSE_END>

3. Generate tokens

Rotation Tokens



1. Extract (X, Y) coordinates

OrientAnything
azimuth: 222
(180 - 222)%360 = 318 ≈ 315°
azimuth: 267
(180 - 267)%360 = 273 ≈ 270°

2. Extract orientation

Token sequence:
<OBJECT_START> <OBJ_PERSON>
<X_-4> <Y_6>
<AZIMUTH_315> <OBJECT_END>
<OBJECT_START> <OBJ_FURNITURE>
<X_-98> <Y_38>
<AZIMUTH_270> <OBJECT_END>

3. Generate tokens

(c)

Atomic

Q: Estimate this person's pose.

A: <POSE_START> <SHOULDER_L> <X_172> <Y_53> <SHOULDER_R> ... <ORIENT_START> <YAW_6> <TORSO_W_2> <ORIENT_END> <POSE_END>

Chain of Thought

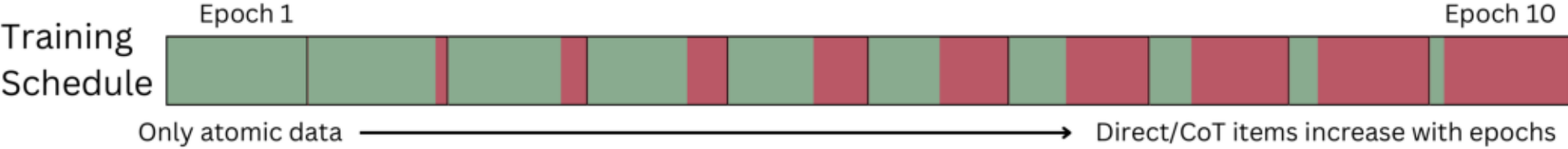
Q: A person and a chair are shown in the image. Which side of the person is the chair on from the person's perspective? To answer this question, let's think through it step by step...

A:

Direct

Q: A person and a chair are shown in the image. Which side of the person is the chair on from the person's perspective?

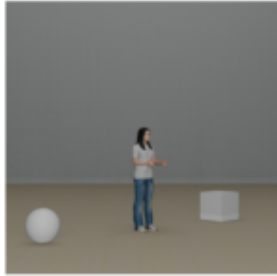
A: Left



(d)

Synthetic

Naturalistic



Which side of the person is the cube on from the person's perspective?



Which side of the person is the bike on from the person's perspective?



Isle Dataset
Goral et al. 2025

3DSRBench

Ma et al. 2025



From the perspective of the humanoid minifigure, where is the object located relative to it?



Orientation
Q: Is the stop sign on the left or right side of the man on the bicycle?
A: On the right side.